T5

$\xi \sim p(x) = \frac{1}{\theta}\{[\theta; 2\theta]\} \Rightarrow F(x) = \left(\frac{x}{\theta} - 1\right)\{[\theta; 2\theta]\}$

$M\xi = \int_{-\infty}^{+\infty} x \cdot p(x)\,dx = \frac{1}{\theta}\int_{\theta}^{2\theta} x\,dx = \frac{1}{\theta}\cdot\frac{1}{2}(4\theta^2 - \theta^2) = \frac{3}{2}\theta$

$M\xi^2 = \int_{-\infty}^{+\infty} x^2 p(x)\,dx = \frac{1}{\theta}\int_{\theta}^{2\theta} x^2\,dx = \frac{1}{\theta}\cdot\frac{1}{3}(8\theta^3 - \theta^3) = \frac{7}{3}\theta^2$

$D\xi = M\xi^2 - (M\xi)^2 = \frac{7}{3}\theta^2 - \frac{9}{4}\theta^2 = \frac{1}{12}\theta^2$

a) ОММ.

$\alpha_1(\theta) = \hat{\alpha}_1 \Rightarrow \frac{3}{2}\theta = \overline{X} \Rightarrow \hat{\theta}_1 = \frac{2}{3}\overline{X}$

ОМП

$L(\theta) = \left(\frac{1}{\theta}\right)^n = \frac{1}{\theta^n}$

$\theta \le x_i \le 2\theta \Rightarrow \theta \le \min(x_i) = X_{(1)} \quad X_{(n)} = \max(x_i) \le 2\theta \Rightarrow$

$\Rightarrow \frac{X_{(n)}}{2} \le \theta \le X_{(1)}$

$L(\theta) \to \max$ при $\theta - \min \Rightarrow \hat{\theta}_2 = \frac{X_{(n)}}{2} = \frac{X_{max}}{2}$

b) $M\hat{\theta}_1 = \frac{2}{3}M\overline{X} = \frac{2}{3}M\xi = \frac{2}{3}\cdot\frac{3}{2}\cdot\theta = \theta \Rightarrow \hat{\theta}_1$ - несмещённая

$D\hat{\theta}_1 = \frac{4}{9}\cdot D\overline{X} = \frac{4}{9n}D\xi = \frac{4}{9n}\cdot\frac{\theta^2}{12} = \frac{1}{27n}\theta^2 \xrightarrow[n\to\infty]{} 0 \Rightarrow$

$\Rightarrow \hat{\theta}_1$ - состоятельна.

$M\hat{\theta}_2 = \frac{1}{2}MX_{max} = \frac{1}{2}\int_{\theta}^{2\theta} x\cdot n\left(\frac{x}{\theta} - 1\right)^{n-1}\cdot\frac{1}{\theta}\,dx = \left\{\begin{matrix} t = \frac{x-\theta}{\theta} & x = \theta(t+1) \\ dx = \theta\,dt & \end{matrix}\right\} =$

$= \frac{n}{2}\int_0^1 \theta(t+1)t^{n-1}\,dt = \frac{n\theta}{2}\left(\int_0^1 t^n\,dt + \int_0^1 t^{n-1}\,dt\right) =$

$= \frac{n\theta}{2}\left(\frac{t^{n+1}}{n+1}\Big|_0^1 + \frac{t^n}{n}\Big|_0^1\right) = \frac{n\theta}{2}\left(\frac{1}{n+1} + \frac{1}{n}\right) = \frac{n\theta}{2}\cdot\frac{2n+1}{n(n+1)} =$

$= \frac{\theta}{2}\cdot\frac{2n+1}{n+1} = \theta\left(1 - \frac{1}{2(n+1)}\right) \Rightarrow \hat{\theta}_2$ - смещённая $\Rightarrow$

$\Rightarrow \hat{\theta}_2' = \frac{n+1}{2n+1}\cdot X_{max}$

$MX_{max}^2 = \int_{\theta}^{2\theta} x^2 n\left(\frac{x}{\theta} - 1\right)^{n-1}\frac{1}{\theta}\,dx = \left\{\begin{matrix} t = \frac{x-\theta}{\theta} & dx = \theta\,dt \\ x = \theta(t+1) & \end{matrix}\right\} =$

$= n\int_0^1 (1+t)^2\theta^2 t^{n-1}\,dt = n\theta^2\int_0^1 (1 + 2t + t^2)t^{n-1}\,dt =$

$$= n\theta^2\left(\int_0^1 t^{n-1}dt + 2\int_0^1 t^n dt + \int_0^1 t^{n+1}dt\right) =$$

$$= n\theta^2\left(\frac{1}{n}\cdot t^n\Big|_0^1 + 2\,\frac{1}{n+1}\cdot t^{n+1}\Big|_0^1 + \frac{1}{n+2}\cdot t^{n+2}\Big|_0^1\right) =$$

$$= n\theta^2\left(\frac{1}{n} + \frac{2}{n+1} + \frac{1}{n+2}\right) = n\theta^2\,\frac{(n+1)(n+2)+2n(n+2)+n(n+1)}{n(n+1)(n+2)} =$$

$$= \theta^2\cdot\frac{4n^2+8n+2}{(n+1)(n+2)}$$

$$D x_{max} = M x_{max}^2 - \left(M x_{max}\right)^2 = \theta^2\left(\frac{4n^2+8n+2}{(n+1)(n+2)} - \frac{(2n+1)^2}{(n+1)^2}\right) =$$

$$= \theta^2\,\frac{n}{(n+1)^2(n+2)}$$

$$D\tilde{\theta}_2' = \left(\frac{n+1}{2n+1}\right)^2 D x_{max} = \left(\frac{n+1}{2n+1}\right)^2\frac{n\theta^2}{(n+1)^2(n+2)} = \frac{n\theta^2}{(n+2)(2n+1)^2} \to$$

$$\xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_2' \text{ – состоятельна.}$$

$$D\tilde{\theta}_2 = \frac{1}{4}D x_{max} = \frac{n\theta^2}{4(n+1)^2(n+2)} \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_2 \text{ – состоятельна}$$

c) $\tilde{\theta}_1 = \frac{2}{3}\overline{x}$ $\qquad$ $\tilde{\theta}_2' = \frac{n+1}{2n+1}x_{max}$

$D\tilde{\theta}_1 = \frac{\theta^2}{27n}$ $\qquad$ $D\tilde{\theta}_2' = \frac{n\theta^2}{(n+2)(2n+1)^2}$

$$\frac{n\theta^2}{(n+2)(2n+1)^2} < \frac{\theta^2}{27n}$$

$$\frac{n}{(n+2)(2n+1)^2} < \frac{1}{27n}$$

$$(n+2)(2n+1)^2 > 27n^2$$

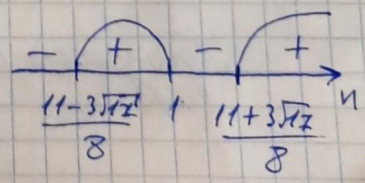
$$(n+2)(4n^2+4n+1) > 27n^2$$

$$4n^3+12n^2+9n+2 > 27n^2$$

$$4n^3-15n^2+9n+2 > 0$$

$$(n-1)(4n^2-11n-2) > 0$$

$$n_{1,2} = \frac{11\pm\sqrt{121+32}}{8} = \frac{11\pm3\sqrt{17}}{8} \approx 3{,}0$$



$\Rightarrow D\hat{\theta}_1 \geq D\hat{\theta}_2'$ при $n \geq 3 \Rightarrow \hat{\theta}_2'$ - эффективнее $\hat{\theta}_1$

d) $f(\vec{x}_n, h) = \frac{x_{max}}{\theta} - 1$

$$P(f \leq t) = P(x_{max} < \theta(t+1)) = (F(\theta(t+1)))^n = \begin{cases} 0, & t \leq 0 \\ t^n, & t \in [0;1] \\ 1, & t \geq 1 \end{cases}$$

$t_1 : F(t_1) = \frac{1-\beta}{2}$

$t_2 : F(t_2) = \frac{1+\beta}{2}$ $\Rightarrow P\left(\sqrt[n]{\frac{1-\beta}{2}} < \frac{x_{max}}{\theta} - 1 < \sqrt[n]{\frac{1+\beta}{2}}\right) = \beta$

$$1 + \sqrt[n]{\frac{1-\beta}{2}} < \frac{x_{max}}{\theta} < 1 + \sqrt[n]{\frac{1+\beta}{2}} \Rightarrow \frac{x_{max}}{1+\sqrt[n]{\frac{1+\beta}{2}}} < \theta < \frac{x_{max}}{1+\sqrt[n]{\frac{1-\beta}{2}}}$$

e) <u>По ОММ:</u>

$$\hat{\theta}_1 = \frac{2}{3}\overline{x} = \frac{2}{3}\hat{\alpha}_1 = g(\hat{\alpha})$$

$$\sqrt{n}\,\frac{g(\hat{\alpha}) - g(\alpha)}{\sigma(\hat{\alpha})} \rightsquigarrow N(0;1) \qquad \sigma^2(\alpha) = \nabla^T g(\alpha) K \nabla g(\alpha)$$

$$\nabla g(\alpha) = \frac{2}{3} \qquad K = \alpha_{1+1} - \alpha_1 \cdot \alpha_1 = \alpha_2 - \alpha_1^2$$

$$\sqrt{n}\,\frac{\hat{\theta}_1 - \theta}{\frac{2}{3}\sqrt{\hat{\alpha}_2 - \hat{\alpha}_1^2}} \rightsquigarrow N(0;1)$$

$$t_1 = U_{\frac{1-\beta}{2}} \qquad t_2 = U_{\frac{1+\beta}{2}}$$

$$t_1 < \sqrt{n}\,\frac{\hat{\theta}_1 - \theta}{\frac{2}{3}\sqrt{\hat{\alpha}_2 - \hat{\alpha}_1^2}} < t_2$$

$$\hat{\theta}_1 - t_2 \cdot \frac{2\sqrt{\hat{\alpha}_2 - \hat{\alpha}_1^2}}{3\sqrt{n}} < \theta < \hat{\theta}_1 - t_1 \cdot \frac{2\sqrt{\hat{\alpha}_2 - \hat{\alpha}_1^2}}{3\sqrt{n}}$$

Проверим на регулярность:

$$\int_\theta^{2\theta} \frac{\partial p(x,\theta)}{\partial \theta}\,dx = \int_\theta^{2\theta} -\frac{1}{\theta^2}\,dx = -\frac{1}{\theta^2}(2\theta - \theta) = -\frac{1}{\theta} \neq 0 \Rightarrow$$

$\Rightarrow$ модель не регулярна.

Продолжение в Т5.ipynb